

Introduction to Biology: Cell Foundations

Cells are the basic structural and functional units of life. Cell theory states that all living organisms consist of one or more cells, that the cell is the basic unit of life, and that all cells arise from existing cells. Prokaryotic cells do not have a membrane-bound nucleus. Eukaryotic cells have a membrane-bound nucleus that houses DNA. The plasma membrane is a selectively permeable boundary made primarily of a phospholipid bilayer. It controls the movement of substances into and out of the cell. The cytoplasm includes the fluid and cellular structures between the plasma membrane and nucleus.

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Cell Organelles and Their Roles

The nucleus stores genetic information in DNA and helps regulate cell activities. Ribosomes synthesize proteins by translating messenger RNA. Mitochondria carry out much of cellular respiration, producing ATP that supplies usable energy for cell processes. The rough endoplasmic reticulum is studded with ribosomes and helps process proteins. The Golgi apparatus modifies, sorts, and packages proteins for transport. Lysosomes contain enzymes that break down waste materials. In plant cells, chloroplasts carry out photosynthesis, and a cellulose cell wall provides structural support.

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Movement Across Membranes

Diffusion is the net movement of particles from a region of higher concentration to lower concentration. Osmosis is the movement of water across a selectively permeable membrane toward a region of higher solute concentration. Passive transport does not require direct cellular energy input and moves substances down their concentration gradient. Active transport requires energy, often ATP, to move substances against their concentration gradient. In a hypotonic solution, water tends to enter an animal cell; in a hypertonic solution, water tends to leave it. In an isotonic solution, there is no net movement of water.

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